



$$\bar{\mathbf{J}}_1 = \mathbf{M}^{-1} \mathbf{J}_1^\top (\mathbf{J}_1 \mathbf{M}^{-1} \mathbf{J}_1^\top)^{-1}, \quad \mathbf{N} = \mathbf{I} - \bar{\mathbf{J}}_1 \mathbf{J}_1$$

$$\tau = \mathbf{J}_1^\top \mathbf{F}_1 + \mathbf{N}^\top \tau_2$$